

International Cybersecurity and Digital Forensics Academy (ICDFA)

CIP-B103: Network Forensics Fundamentals

LAB 7: DNS Introduction and Traffic Analysis

Three-Week Delivery Block 3/3 | 1-2 August 2026

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Important Pre-Lab Instruction Read the matching lecture slide deck before executing the practical. Work only inside the ICDFA-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.	
Course Code	CIP-B103
Course Title	Network Forensics Fundamentals
Lab Number	Lab 7
Lab Title	DNS Introduction and Traffic Analysis
Delivery Block	3/3 of 3
Scheduled Dates	1-2 August 2026
Estimated Duration	3-4 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	dig_dns.pcap or locally generated DNS captures; optional smtp.pcap correlation

1. Source Materials and Slide Alignment

- Domain Name System Forensics: domain-to-IP resolution, hierarchical DNS, resolver/TLD/authoritative roles, dig queries, capture with tshark, query/response fields and DNS within email traffic.
- The slide practical captures dig and browser DNS traffic, examines A records, local resolver configuration, TTL and authoritative information.
- The exercise builds the normal DNS baseline required before analyzing spoofing in Lab 8.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded CIP-B103 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

You are documenting normal DNS behaviour for an incident-response baseline. You will identify the configured resolver, capture controlled dig queries, analyze transaction IDs, names, record types, response codes, answers and TTL values, and correlate DNS resolution with a subsequent web or email connection.

3. Learning Outcomes

- Explain recursive and iterative DNS resolution at a foundational level.
- Use dig to query A, AAAA, MX and NS records.
- Capture and analyze DNS queries and responses with tshark/Wireshark.
- Identify transaction ID, query name, record type, response code, answer and TTL.
- Correlate DNS responses with later IP connections.
- Recognize normal variations such as multiple answers, CNAMEs, cache responses and IPv6 queries.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.

- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.

5. Required Tools and Environment

Item	Recommended Tool	Purpose
Operating system	Kali Linux or Ubuntu VM	Generate and analyze DNS traffic.
DNS utility	dig from dnsutils	Perform controlled queries.
Capture tools	tshark and Wireshark	Capture DNS traffic.
Evidence	dig_dns.pcap or new PCAPNG	Analysis source.
Resolver config	/etc/resolv.conf, resolvectl	Identify configured DNS service.
Optional browser	Firefox/Chrome	Generate multiple DNS lookups.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/CIP-B103-Lab7/{evidence,working,exported,reports,screenshots,scripts}
cd ~/CIP-B103-Lab7
pwd
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

```
sudo apt update
sudo apt install -y dnsutils tshark wireshark
cat /etc/resolv.conf | tee reports/resolv_conf.txt
resolvectl status 2>/dev/null | tee reports/resolvectl_status.txt || true
wget -O evidence/dig_dns.pcap \
  'https://raw.githubusercontent.com/frankwxu/digital-forensics-
  lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap'
cp --preserve=timestamps evidence/dig_dns.pcap working/dig_dns_working.pcap
sha256sum evidence/dig_dns.pcap working/dig_dns_working.pcap | tee reports/dns_capture_hashes.txt
```

Screenshot required: Configured resolver information, evidence details and hashes.

7. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	CIP-B103-Lab7-YourFullName
Trainee name	
Date and time started	
Evidence file name(s)	
Source or generation method	
Original SHA-256	
Working-copy SHA-256	
Analysis workstation/VM	
Notes on any changes	

8. Part A - Query DNS Records with dig

```
dig example.com A | tee reports/dig_example_A.txt
dig example.com AAAA | tee reports/dig_example_AAAA.txt
dig example.com MX | tee reports/dig_example_MX.txt
dig example.com NS | tee reports/dig_example_NS.txt
dig +short example.com A | tee reports/dig_example_short.txt
```

- Record the resolver shown in the SERVER line.
- Record status/response code, flags, question count, answer count and query time.
- Explain IN, A, AAAA, MX and NS.
- Record at least one TTL and explain that it is a cache lifetime, not a creation date.

9. Part B - Capture a Fresh DNS Query

```
IFACE=eth0
sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w evidence/fresh_dig_dns.pcapng &
sleep 3
dig +noedns example.com A >/dev/null
wait
sha256sum evidence/fresh_dig_dns.pcapng | tee reports/fresh_dns_sha256.txt
```

Interface Note

On systems using a local stub resolver, packets may appear on lo and then on a physical interface. Use tshark -D to list interfaces and choose the one that shows the actual query.

10. Part C - Extract DNS Query and Response Fields

```
PCAP=evidence/fresh_dig_dns.pcapng
# Query
tshark -r "$PCAP" -Y 'dns.flags.response==0' -T fields \
  -e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
  -e dns.id -e dns.qry.name -e dns.qry.type \
  | tee reports/dns_queries.tsv

# Response
tshark -r "$PCAP" -Y 'dns.flags.response==1' -T fields \
  -e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
  -e dns.id -e dns.flags.rcode -e dns.count.answers -e dns.a -e dns.aaaa -e dns.resp.ttl \
  | tee reports/dns_responses.tsv
```

11. Part D - Match Queries to Responses

Transaction ID	Query Name	Type	Client/Resolver	Response Code	Answer(s)	TTL	Time Delta

- Match query and response using transaction ID plus the endpoint tuple.
- Confirm that the response source is the queried resolver.
- Calculate response time from frame timestamps.
- Document CNAME chains and multiple A/AAAA answers where present.

12. Part E - Analyze DNS Generated by a Browser

A browser may request several domains for the page, images, fonts, telemetry or cached services. Capture a visit to a non-sensitive instructor-approved site and inventory all query names.

```

sudo tshark -i IFACE -f 'port 53' -a duration:40 -w evidence/browser_dns.pcapng &
sleep 3
# Open the approved site in a private browser window, then wait.
wait

tshark -r evidence/browser_dns.pcapng -Y 'dns.flags.response==0' -T fields -e dns.qry.name -e dns.qry.type \
| sort | uniq -c | sort -nr | tee reports/browser_dns_inventory.txt

```

13. Part F - Correlate DNS with Subsequent Connections

```

# Extract DNS A answers
tshark -r evidence/browser_dns.pcapng -Y 'dns.a' -T fields -e frame.time_epoch -e dns.qry.name -e dns.a \
| tee reports/dns_A_answers.tsv

# Extract later TCP destinations
tshark -r evidence/browser_dns.pcapng -Y 'tcp.flags.syn==1 && tcp.flags.ack==0' -T fields \
-e frame.time_epoch -e ip.dst -e tcp.dstport \
| tee reports/subsequent_tcp_destinations.tsv

```

Select one DNS answer and show whether the client later initiates a connection to that IP. Explain any mismatch caused by multiple answers, proxies, CDNs, IPv6, caching or capture duration.

14. Part G - DNS Analysis Within SMTP Evidence

If smtp.pcap from Lab 4 is available, filter dns and identify the query used to locate a mail server. Record the queried name, record type, resolver, answer and TTL.

```

# Optional, adjust path to your Lab 4 capture
tshark -r ../CIP-B103-Lab4/working/smtp_working.pcap -Y 'dns' -T fields \
-e frame.number -e frame.time -e dns.qry.name -e dns.qry.type -e dns.a -e dns.resp.name -e dns.resp.ttl \
| tee reports/smtp_dns_correlation.tsv

```

15. Required Findings Worksheet

Field	Finding
Configured resolver IP	
Client source port	
Resolver destination port	
Transaction ID	
Query name and type	
Response code	
Answer IP(s)	
TTL	
Query-response time delta	
Subsequent connection correlation	
Normal variations observed	

Required Screenshots Checklist

- ☐ Resolver configuration.
- ☐ dig A/AAAA/MX/NS outputs.
- ☐ Fresh DNS capture hash.
- ☐ DNS query fields.

- ☐ Matching DNS response fields.
- ☐ Transaction ID correlation.
- ☐ Answer and TTL.
- ☐ Browser query inventory.
- ☐ DNS answer to TCP destination correlation.
- ☐ Optional SMTP DNS evidence.

Student Report Template

1. Executive summary.
2. Resolver and environment.
3. Evidence integrity.
4. dig results.
5. Packet-level query/response analysis.
6. Transaction matching and timing.
7. Browser DNS inventory.
8. Connection correlation.
9. Limitations and normal variations.
10. Appendix.

Submission Requirements

- One PDF report named CIP-B103-Lab7_RegNo_FullName.pdf.
- The original or generated PCAP/PCAPNG evidence file and its SHA-256 hash text file.
- A reports folder containing exported CSV/TXT command output and any recovered objects.
- Screenshots must be readable, numbered and referenced in the report narrative.
- Compress the report and evidence outputs into one ZIP named with your registration number and lab number.
- Do not submit passwords or decoded credentials in public discussion channels; include them only in the protected assessment submission where required.

Marking Rubric (100 Marks)

Assessment Area	Marks	What the Instructor Will Check
Preparation and resolver identification	10	Correct resolver and evidence hashes.
dig record analysis	15	A, AAAA, MX, NS and TTL interpretation.
Packet-field extraction	25	Correct IDs, names, types, endpoints and response codes.
Query-response correlation	20	Accurate matching and timing.
Browser/connection correlation	20	Inventory and defensible IP connection linkage.
Report quality	10	Clear screenshots, tables and conclusions.

Instructor Quick Answer Guide

Checkpoint	Expected Observation
Standard transport	Usually UDP destination port 53; TCP may be used for large responses or zone transfer.
Query flag	dns.flags.response==0.
Response flag	dns.flags.response==1.
A record	IPv4 address.
AAAA record	IPv6 address.
TTL	How long a resolver may cache the record.

Command Reference Summary

Command	Purpose
dig NAME TYPE	Query a DNS record.
dns.qry.name	Queried domain name.
dns.qry.type	Requested record type.
dns.id	DNS transaction identifier.
dns.flags.rcode	Response code.
dns.resp.ttl	Answer TTL.

Academic Integrity and Authorization Statement

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.